

Rihua Van Steenburgh

Data Engineer | Cloud Data Platforms | AI Data Systems

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Summary

Data Engineer with hands-on project experience building cloud data pipelines and lakehouse workloads across Azure and AWS, plus analytics-ready data models and AI data systems. Skilled in Python, SQL, Databricks, PySpark, dbt, PostgreSQL/pgvector, Docker, and CI/CD, with experience in API ingestion, ETL/ELT, dimensional modeling, data quality, hybrid retrieval, and RAG evaluation.

Skills

- **Languages:** Python, SQL, R, JavaScript, TypeScript
- **Cloud & Data Platforms:** Azure Data Factory, ADLS Gen2, Databricks, AWS S3, ECS/Fargate, Redshift Serverless, EventBridge, CloudWatch
- **Data Engineering:** API ingestion, ETL/ELT, batch pipelines, orchestration, data quality, CI/CD, dimensional modeling, star schema
- **Processing & Analytics:** PySpark, Delta Lake, dbt, PostgreSQL, Power BI
- **AI / Retrieval:** pgvector, embeddings, vector search, hybrid retrieval, RRF, RAG, retrieval evaluation
- **Development & DevOps:** FastAPI, Docker, Airflow, GitHub Actions, Git/GitHub, Streamlit, Next.js
- **ML / NLP:** scikit-learn, TF-IDF, Linear SVM, PyTorch, Transformers, DistilBERT

Projects

1. NYC 311 Service Requests Lakehouse | Azure • Databricks • PySpark • SQL • Delta Lake

Mar 2026 - May 2026 [GitHub](#) | [Case Study](#)

- Built an Azure lakehouse pipeline using Azure Data Factory, ADLS Gen2, Databricks, Python/PySpark, SQL, and Delta Lake to process Bronze, Silver, and Gold layers.
- Implemented reusable ingestion, cleaning, SQL validation, and star-schema modeling to deliver analytics-ready fact tables, Gold marts, runbooks, architecture notes, and cloud execution proof.

2. Cloud Flight Fare Pipeline | AWS • Python • SQL • ECS/Fargate • S3 • Redshift • dbt

Feb 2026 - May 2026 [GitHub](#) | [Case Study](#)

- Built an end-to-end AWS batch pipeline using Python, Docker, ECS/Fargate, EventBridge Scheduler, S3, Redshift Serverless, SQL, and dbt to ingest and transform flight fare API data.
- Developed dbt staging models, dimensional marts, data quality tests, CI checks, architecture diagrams, runbooks, CloudWatch proof logs, and cost/secret safety documentation.

3. CivicLens RAG Operations Copilot | Python • FastAPI • PostgreSQL/pgvector • Hybrid Retrieval • Docker

May 2026 - Aug 2026 [GitHub](#) | [Live Demo](#) | [Case Study](#)

- Engineered a RAG ingestion and indexing pipeline with manifest validation, section-aware chunking, stable IDs, SQL migrations, and PostgreSQL/pgvector; implemented hybrid semantic and full-text retrieval with RRF.
- Deployed a Next.js → Render FastAPI → Neon PostgreSQL/pgvector application with Docker, citation validation, safe abstention, and CI; achieved **83.9% Recall@5** and **92.9% expected-source retrieval**.

4. Financial Complaint Auto-Routing with NLP | Python • scikit-learn • TF-IDF • Linear SVM • PyTorch • DistilBERT

Jun 2026 - Aug 2026 [GitHub](#) | [Case Study](#)

- Built a leakage-safe eight-class CFPB complaint-routing study comparing a locked TF-IDF + Linear SVM benchmark with a frozen DistilBERT challenger; used group-aware 2024 splits with zero normalized-text overlap.
- Improved shared-2024 Macro F1 from 0.7671 to 0.7949 and evaluated both frozen models on a 30,156-row retrospective 2025 cohort using model-specific Auto-Route/Human Review policies.

Experience

IT Intern - Fulton County | Jan 2024 - Apr 2024

- Standardized IT asset and network inventory records, improving data consistency and audit readiness.
- Reconciled records across multiple systems to support inventory reporting and internal documentation.

Floral Assistant - Kroger | Dec 2022 - Aug 2023

- Reconciled inventory, pricing, and shrink records to improve reporting accuracy and exception tracking.

Education

- **Master of Science (M.S.) in Information Technology, Data Analytics Concentration** - Middle Georgia State University, Expected Dec 2026 | GPA: 4.00
- **Associate of Applied Science (A.A.S.) in Website Design/Development** - Gwinnett Technical College, Dec 2024 | GPA: 3.96
- **Bachelor of Arts (B.A.) in Chinese Ethnic Language and Literature** - Guangxi University for Nationalities, China